GaitEval: A Multi-Condition Benchmark for Evaluating Whether Large Language Models Use Pose-Derived Gait Evidence

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Abstract

Pose-based gait data are widely used for clinical screening, yet whether Large Language Models (LLMs) can interpret such motion evidence remains underexplored. We introduce GaitEval, a six-dataset benchmark for evidence-grounded label prediction across ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT, with heterogeneous populations, pose formats, and label spaces. A baseline-anchored protocol compares controlled pose-derived inputs with label-prior and same-feature linear references, and evaluates zero-shot, few-shot, and low-rank adaptation (LoRA) within their tested coverage. The retained five-family motion-summary evaluation shows near-chance agreement on its original tasks; a shared six-dataset, subject-disjoint Qwen2.5-7B evaluation identifies important dataset–representation exceptions. Linear references recover task-relevant signal where available, while adaptation recovers some in-distribution mappings: standardized GMFCS summary macro-F1 rises from 0.265 to 0.467, but remains below its matched linear reference of 0.579. The accepted-protocol ATAXIA result likewise retains its agreement increase from $\kappa = 0.046$ *to* 0.507*; transfer remains unreliable in the tested bidirectional ATAXIA–PATHOGAIT probe. Overall, evidence use is heterogeneous and representation-dependent: adaptation can learn dataset-specific mappings, while reliability beyond the tested settings remains unresolved.*

1. Introduction

Human gait provides an accessible signal for assessing balance, coordination, strength, and mobility [1–3]. Advances in video-based pose estimation allow ordinary walking videos to be converted into skeleton trajectories without specialized motion-capture systems [4–6]. Clinical reviews describe how this low-cost route has been used across neurological and rehabilitation settings [7, 8]. Pose trajectories provide a compact, anatomically indexed representation from which interpretable measurements of motion, symmetry, stability, and variability can be derived [9, 10].

Most pose-based clinical gait methods are task-specific supervised systems that map skeleton sequences to gait classes, severity levels, or clinical scores [11, 12]. Clinical studies have related quantitative gait measurements to motor severity and function in cerebral palsy, Parkinsonian impairment, and ataxia [11, 13, 14]. Related visual studies span heterogeneous cohorts, acquisition settings, and label definitions [11, 12, 15], complicating direct cross-resource comparisons. Thus, cross-condition evaluation is not itself the novelty of this work. Instead, we ask whether LLM predictions are grounded in supplied pose-derived evidence. This distinction is important because many gait resources contain numerous clips from relatively few participants, requiring participant-aware interpretation.

Pose evidence can be presented as raw coordinates, deterministic motion summaries, or biomechanical features. Poor performance may therefore reflect limited task-relevant signal, difficulty interpreting the representation, or failure to map informative evidence to dataset-specific labels. Accuracy alone cannot distinguish these cases, particularly for imbalanced ordinal and multi-class tasks [16, 17]. Evaluation should therefore compare LLMs with label-prior baselines and simple classifiers trained on the same evidence.

We introduce **GaitEval**, a multi-condition benchmark in which *gait reasoning* means grounding dataset-specified label predictions in pose-derived evidence, rather than open-ended diagnosis or clinical interpretation. The benchmark comprises six datasets: ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT. Each retains its population provenance and native label space. GaitEval introduces no new clinical recordings; it provides controlled representations, evidence-use tests, and same-feature diagnostic references (Fig. 1).

The evaluation follows the original evidence ladder: zero-shot prompting, few-shot examples, and LoRA adaptation, with transfer limited to the tested ATAXIA–PATHOGAIT pair [18, 19]. The common six-dataset analysis uses Qwen2.5-7B-Instruct, subject-disjoint partitions, and capture-quality-free features. Additional model, view, and adaptation measurements retain their original protocol labels and are never substituted for missing common-protocol experiments.

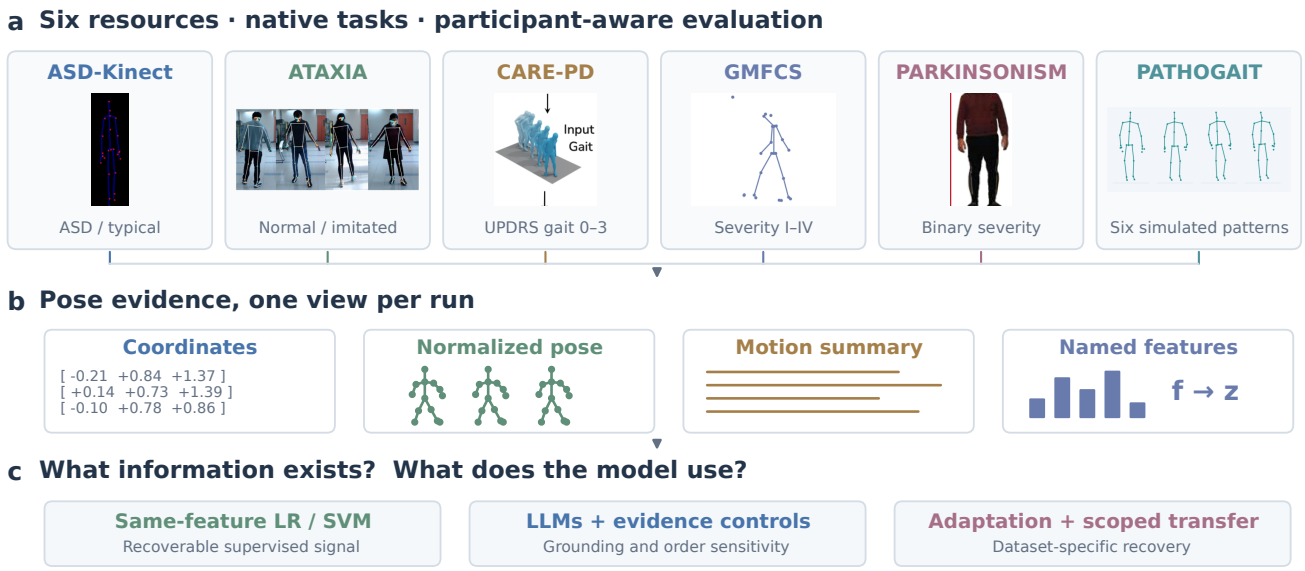


Figure 1. **GaitEval overview.** Six datasets enter one evidence-grounded benchmark. Source-publication examples and recorded-skeleton renderings illustrate the resources; they are not matched test cases or inputs to an image model. One pose-derived evidence view is supplied per run. Same-feature references test recoverable signal; evidence/order controls, adaptation, and scoped transfer test the limits of LLM use. Source attribution and experiment coverage are documented in the supplement.

Our main contributions are:

- We introduce GaitEval, a six-dataset benchmark for testing whether LLM predictions are grounded in pose-derived gait evidence across heterogeneous resources, representations, and label spaces.
- We provide a baseline-anchored protocol that separates limited input signal from failed evidence use by comparing controlled LLM inputs with label-prior and linear feature-based references.
- We evaluate an evidence ladder from zero-shot prompting to few-shot adaptation and LoRA fine-tuning within its measured coverage, showing in-distribution recovery on selected tasks but no reliable transfer on the tested compatible dataset pair.

2. Related Work

2.1. Clinical Gait Analysis and Pose-Based Representations

Clinical gait analysis supports assessment of motor impairment, functional severity, and rehabilitation outcomes [1, 2, 20]. Video-based pose estimation offers an accessible alternative to laboratory motion capture by converting ordinary recordings into joint trajectories [7, 8]. Markerless toolkits and smartphone-based pipelines further broaden access to quantitative movement analysis [21, 22]. Validation studies have compared video-derived temporal and kinematic measures with instrumented walkways or motion-capture systems across different gait patterns and clinical cohorts [23–25]. These studies also show that viewpoint, occlusion, pose-estimation error, and population differences remain important sources of measurement uncertainty.

Gait resources differ in population, acquisition setup, skeleton format, and label definition [11, 13, 15]. Pose data provide a shared semantic interface, but do not make these resources clinically equivalent. GaitEval therefore preserves each resource's native representation and dataset-defined task.

2.2. Visual and Pose-Based Pathological Gait Analysis

Pose-based gait systems use biomechanical features or learned temporal representations for supervised prediction. Graph models such as ST-GCN, 2s-AGCN, and MS-G3D are widely used for skeleton analysis [26–28], but are generally tied to fixed datasets, skeletons, and label spaces.

Early low-cost RGB and RGB-D systems measured or classified pathological gait using semantic gait features, silhouettes, or tracked skeletons [29–31]. Parkinsonian assessment has since ranged from single-camera kinematic measurement and natural-gait monitoring to learned skeleton–silhouette models for clinical severity estimation [32–34]. Related work has also estimated cerebral-palsy gait function and kinematics and quantified gait after stroke from ordinary videos [13, 35, 36].

For ataxia, Auto-Gait analyzes clinically recorded gait-task videos from multiple sites, while pose-based work has also evaluated performed ataxic gait [11, 37]. A separate

clinical study used video keypoints to distinguish Parkinson's disease from spinocerebellar degeneration, showing that cross-condition visual analysis predates GaitEval [38]. Broader abnormal-gait resources include simulated pathological patterns and clinically annotated abnormal-gait videos [15, 39]. Collectively, these studies establish that visual and pose features can support condition-specific prediction; results from performed or simulated gait, however, do not establish patient-level diagnostic validity.

Cross-dataset gait analysis is also established. Transfer-learning and spatiotemporal classifiers have been evaluated across separately collected pathological-gait datasets [40, 41]. Auto-Gait includes leave-one-site-out evaluation, and CARE-PD formalizes within-dataset, cross-dataset, leave-one-dataset-out, and adaptation protocols [37, 42]. GaitEval therefore does not claim domain-shift difficulty as a new finding. Instead, it tests whether general-purpose LLMs use supplied pose evidence under same-input controls. Logistic regression and a linear support vector machine receive the same deterministic features, and the ATAXIA–PATHOGAIT transfer experiment is limited to that compatible dataset pair.

2.3. Language Models for Human Motion and Clinical Gait

Motion-language systems such as MotionGPT, MotionLLM, and MoChat connect language models with motion through learned tokenization or cross-modal alignment [43–45]. Gait-specific models have also been developed for biometric recognition and Parkinsonian scoring [46, 47]. BiomechGPT and BioGait-VLM further connect language-aligned representations with clinically relevant movement analysis [48, 49]. These methods rely on task-specific motion-language training rather than testing off-the-shelf LLM evidence use.

The closest concurrent work evaluates general-purpose LLMs using text-encoded kinematic waveforms [50]. More broadly, prior work has examined LLMs for structured and tabular inputs [51] and multimodal instruction following [52, 53]. GaitEval instead evaluates heterogeneous gait resources through controlled pose-derived views, same-feature linear references, and adaptation experiments. Its purpose is to separate limited motion signal from representation difficulty and failed LLM evidence use, while restricting transfer conclusions to the tested dataset pair.

3. Methods

3.1. Source Sequences and Dataset Tasks

GaitEval evaluates dataset-specified label prediction from pose-derived evidence. The six datasets and their common-protocol partitions are listed in Table 1. GMFCS contains children with cerebral palsy [55]; PARKINSONISM and CARE-PD concern Parkinson's disease with different severity or gait-item targets [12, 42]. ATAXIA contains healthy volunteers imitating ataxic gait [56], and PATHOGAIT contains healthy volunteers simulating six gait patterns [15]. ASD-Kinect defines an autism-versus-typical task [67]; its source reports some movements simulated by others without identifying the affected sequences. These resources are therefore not six interchangeable clinical cohorts.

A source sequence is $X_i \in \mathbb{R}^{T_i \times J_d \times C_d}$, with dataset-specific joints and coordinate channels. Native 2D/3D representations follow established pose formats [4, 54]. The retained accepted-protocol measurements (L) use a 124-frame source tensor, with padding masked during feature computation. This tensor is the *source*, not the representation supplied identically to every LLM: normalized-coordinate prompts use 32 sampled frames and 12 common joints, whereas features and summaries aggregate valid-frame trajectories. The common S corpus has its own native/canonical data layers and feature definitions; 124-frame padding and the L coordinate prompt are not imposed on it.

All six S partitions are reported as subject-disjoint in the committed reference manifests. Their GMFCS result replaces the known overlapping legacy split in the main comparisons. L and S differ in preprocessing, splits, feature columns, prompts, and output parsing. Consequently, we compare methods only within a matched contract and never pool L/S scores. The released predictions verify test-record alignment; independent verification of train/test separation requires the original full split manifests, which are not present in this snapshot. This distinction is retained in the release documentation.

3.2. Controlled Input Views

Each run exposes one view. Four input-view families are defined: raw coordinates, normalized coordinates, deterministic motion summaries, and named biomechanical features. A train-statistics z-scored encoding is a scaling control within the feature family, not a fifth modality. Coordinate prompting is a subset experiment; it is not claimed to have common six-dataset coverage.

Features describe motion range, bilateral asymmetry, trunk/pelvis variability, trajectory variance, and jerk-based smoothness [9, 10]. Unsupported joints are omitted consistently from the LLM and matched linear inputs. These are pose-coordinate descriptors, not calibrated physical stride lengths or forces. The jerk descriptor uses third finite differences per frame and is not asserted to have SI jerk units. For feature f, scaling is $z = (f - \mu_{\mathrm{tr}})/\sigma_{\mathrm{tr}}$, using training statistics only; constant and missing-column handling is specified in the supplement.

Motion summaries are generated by a deterministic, label-independent template from the numerical measurements, not by annotators, dataset metadata, or another LLM. They preserve numbers and fixed qualitative descrip-

Table 1. **Six benchmark datasets under the common subject-disjoint protocol (S).** Counts are eligible train/test clips and test participants, not full-source cohort sizes. All six datasets have summary and z-feature zero-shot, evidence/order controls, summary-SFT, and matched LR/SVM.

Dataset	Population	Task	Train / test	Test subjects
ASD-Kinect	ASD / typical; provenance caveat	Binary	70 / 20	20
ATAXIA	Healthy, imitated gait	Binary	28 / 8	4
CARE-PD	Multi-site Parkinson's cohort	UPDRS gait 0–3	1901 / 665	24
GMFCS	Cerebral-palsy cohort	Four-level ordinal	914 / 255	149
PARKINSONISM	Parkinson's cohort	Binary severity	83 / 29	5
PATHOGAIT	Healthy, simulated gait	Six patterns	5040 / 1440	2

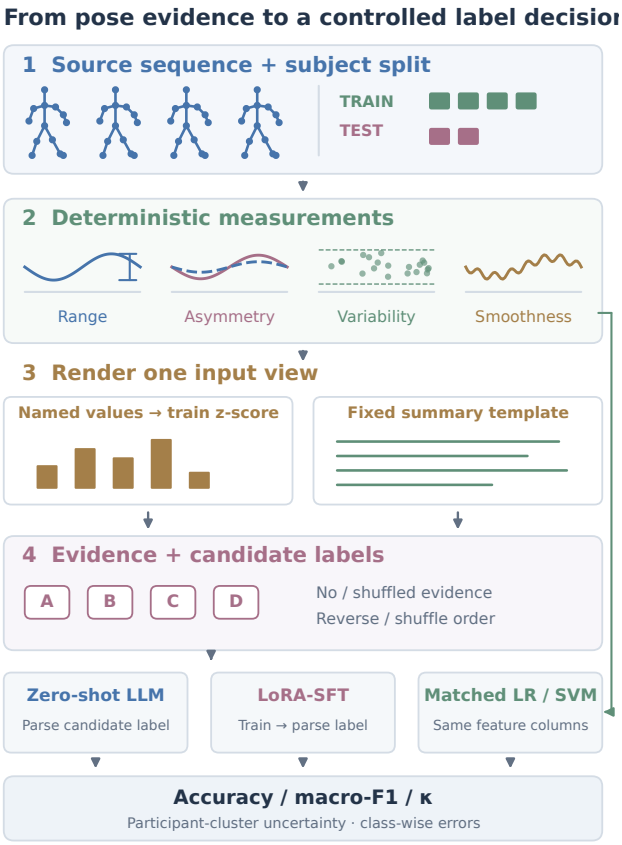


Figure 2. **Pose-to-label evaluation.** Training-only transformations and matched feature references separate signal availability from LLM evidence use. Icons and miniature traces illustrate operations; they are not measured patient trajectories. Optional coordinate, few-shot, feature-SFT, and transfer probes have the coverage reported in the supplement.

tors. Capture-quality variables, including valid-frame fraction, are excluded from every S feature/summary condition. Model-visible inputs exclude answer-revealing labels, dataset names, participant IDs, demographics, and clinical metadata; candidate labels remain because they define the task. Figure 2 distinguishes these transformations.

3.3. Models, Adaptation, and Evidence Controls

The five-family motion-summary axis evaluates Llama-3.3-70B, Qwen2.5-72B, and Gemma-3-27B [57–59], together with GLM-4-32B and DeepSeek-R1-Distill-Qwen-32B [60, 61]. The shared six-dataset axis fixes Qwen2.5-7B-Instruct. The former has mixed L/S coverage and cannot establish a homogeneous six-dataset model ranking; full cells and scope are in the supplement.

All six S datasets use seven common conditions: summary zero-shot; z-feature zero-shot; no evidence; shuffled evidence; reversed and shuffled candidate order; and summary LoRA-SFT. Shuffled evidence deranges summaries across test records while preserving candidate labels. Candidate-order controls preserve evidence and change only label presentation. Decoding is deterministic. The parser chooses the last complete, boundary-anchored candidate mention, resolving same-position matches by length. It does not validate the model's reasoning. Outputs with no matching candidate count as errors.

Few-shot prompting prepends fixed class-balanced training examples [18]. The tested L settings are 8/16/32 examples on ATAXIA, 16 on PARKINSONISM, and 12 on PATHOGAIT; other cells are unexecuted. A reasoning-anchored summary prompt asks for salient motion evidence and candidate comparison [62]; it is also a scoped probe.

LoRA adapts Qwen2.5-7B separately per dataset [19]. The committed S recipe uses rank 16, $\alpha = 32$, dropout 0.05, all linear projections, completion-only loss, three epochs, learning rate 10^{-4}, and effective batch size 16. It saves the final adapter without validation checkpoint selection. No new training is claimed in this revision. Complete prompts, legacy variants, inference budgets, and configuration coverage are given in the supplement.

3.4. References, Metrics, and Scope

Label-prior predictors and class-balanced logistic regression (LR) and linear SVM [63, 64] anchor interpretation. LR/SVM receive the exact deterministic feature columns underlying the corresponding LLM feature view; they are supervised signal probes, not zero-shot competitors or clinical

state of the art.

Accuracy, fixed-label-set macro-F1, and Cohen's κ [16, 17, 65] are reported jointly. Here $\kappa = (p_o - p_e)/(1 - p_e)$, with p_e obtained from the empirical target and prediction marginals. Invalid responses remain errors. Ordinal tasks additionally report quadratic-weighted κ and absolute level error. Macro-F1 uses the complete task label set even when a class is unpredicted.

Participant-cluster bootstrap resampling uses 2,000 draws [66]. Historical bootstrap intervals and paired tail areas are identified separately from fixed-label point metrics, since absent classes and an invalid-output sentinel can change the resampled label set. Very small participant counts limit inference regardless of the number of clips. PATHOGAIT's two-participant results are descriptive; no participant-population superiority is claimed. Multiple exploratory contrasts are reported without a family-wide significance claim.

Transfer is measured only on the L ATAXIA–PATHOGAIT pair after harmonizing normal versus abnormal gait. A model adapted on one dataset is tested on the other without target labeled examples. This probes combined condition and representation shift; no transfer result is claimed for unmatched severity, gait-score, or ASD targets.

4. Experiments

4.1. Shared Six-Dataset Evaluation

The shared S evaluation fixes the same model, input derivation contract, controls, and eligible test records within each dataset. Table 2 reports all six datasets together. The 42 common cells are backed by individual predictions; two extra raw-feature cells and ten five-family cells provide incomplete additional axes. One archived mixed-task CARE-PD diagnostic is excluded from the common task. No unexecuted cell is interpreted as a failure.

Zero-shot evidence use. Summary zero-shot macro-F1 is near stratified chance or below it on several datasets, but performance alone does not establish whether a model used the input. CARE-PD exceeds shuffled evidence by 0.065 macro-F1 (archived participant-cluster 95% interval $[0.007, 0.118]$, paired bootstrap tail area 0.029). This is evidence of clip-specific responsiveness in that tested setting, while the 0.321 score remains far below its 0.659 LR reference. GMFCS's summary advantage over shuffled evidence is 0.052 (tail area 0.089), and reversing candidate order raises macro-F1 to 0.322. The latter sensitivity cautions against equating plausible accuracy with stable evidence grounding.

The five-family axis supports the accepted finding of near-chance L agreement ($\kappa = -0.069$ to 0.035), but these are protocol-scoped results. The available S CARE-PD cells have $\kappa = 0.099$–0.130; ASD-Kinect ranges from -0.100 to 0.200. Thus, neither the original nor expanded evidence justifies a universal claim that zero-shot LLMs cannot use pose-derived evidence. Legacy GMFCS cells affected by unresolved split provenance are retained only for audit in the supplement and are excluded from the main six-dataset comparisons.

4.2. Input Representation and Recoverable Signal

Input views can change prediction distributions without consistently restoring agreement. In the accepted L view experiments, κ remains between -0.095 and 0.057; this range must not be applied to S. Under S, PARKINSONISM's named z-feature view reaches 0.757 macro-F1 and $\kappa = 0.520$, versus 0.396 macro-F1 for summaries. Its archived subject-cluster interval is $[0.543, 0.906]$, based on only 29 clips from five participants. The point estimate exceeds the matched linear references, but this small test does not establish population-level superiority. CARE-PD shows the opposite representation preference: summary macro-F1 is 0.321 versus 0.133 for z-features.

On ASD-Kinect, summary and z-feature scores are 0.310 and 0.436, both below the matched linear point estimates. ATAXIA's eight clips cannot support a powered model ranking. PATHOGAIT is low under both zero-shot views; GMFCS summaries outperform its z-feature encoding. These six rows jointly show representation dependence rather than a universally best input view.

The same-feature references establish recoverable supervised signal, with LR macro-F1 from 0.561 to 0.881 under S. Reference strength and uncertainty remain dataset-specific: ASD-Kinect LR has a wide interval and an archived permutation value of 0.09. They do not show that supervised and zero-shot learning have equivalent information access. Per-class recall and confusion matrices in the supplement expose concentrated outputs that aggregate accuracy can conceal.

4.3. Adaptation Recovers Dataset-Specific Mappings

Summary-SFT changes macro-F1 by 0.000, 0.061, 0.124, 0.202, 0.146, and 0.382 for ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT, respectively. GMFCS improves to 0.467; its archived paired participant-bootstrap tail area is 0.001. CARE-PD improves to 0.445, but its corresponding tail area is 0.43. Both remain below matched LR. PARKINSONISM SFT remains below its zero-shot z-feature result. PATHOGAIT improves substantially but remains below its capture-quality-free linear references; two participants prevent a population-level gain claim. ASD-Kinect shows no summary-SFT macro-F1 gain, despite a change in the class receiving most predictions.

The accepted-protocol adaptation ladder is preserved as a separate, explicitly scoped experiment. Few-shot ATAXIA improves from 0.406 to 0.697 macro-F1. Within the same 7B

Table 2. **Same-model evidence use and adaptation across all six datasets (S).** Values are macro-F1, recomputed from released predictions with a fixed task label set. LR/SVM values are the committed task-matched supervised references. N is clips/participants. Best LLM view is bold; bold does not imply superiority over references.

Dataset	N	Chance	ZS summary	ZS z-features	Summary SFT	LR	SVM
ASD-Kinect	20/20	.490	.310	**.436**	.310	.697	.744
ATAXIA	8/4	.488	.273	**.333**	**.333**	.873	.873
CARE-PD	665/24	.250	.321	.133	**.445**	.659	.650
GMFCS	255/149	.250	.265	.150	**.467**	.579	.571
PARKINSONISM	29/5	.494	.396	**.757**	.542	.561	.574
PATHOGAIT	1440/2	.166	.069	.139	**.451**	.881	.897

model, ATAXIA summary agreement rises from $\kappa = 0.046$ to 0.507, and feature-SFT reaches 0.762 macro-F1 versus LR 0.763. Its participant comparison does not detect a difference ($p = 0.97$); this is not an equivalence test. PATHOGAIT feature-SFT reaches 0.687 versus LR 0.662, but the direction reverses for one of its two participants. These values retain the accepted in-distribution recovery finding; they are not S reruns or evidence of general diagnostic knowledge.

One L PATHOGAIT feature, valid-frame fraction, carries capture-related information. Removing it reduces LR/SVM macro-F1 from 0.662/0.644 to 0.564/0.560. The original feature-SFT was not rerun with that feature removed. The common S evaluation excludes it for all six datasets, but also changes splits and feature construction, so it cannot isolate that single feature's causal contribution. We therefore retain the L adaptation result only as learning from its supplied feature set, and make no kinematics-only superiority claim.

4.4. Transfer and Qualitative Diagnostics

In the tested L ATAXIA–PATHOGAIT transfer probe, in-domain κ is 0.549 and 0.917 for the two directions, while target-domain κ is 0.057 and 0.000. PATHOGAIT is binarized to an approximately 11%/89% target distribution; the resulting binary accuracy cannot be compared with six-class scores. This supports only the accepted conclusion that reliable transfer was not observed on this harmonized pair. Six-dataset membership does not imply task-compatible transfer pairs.

The supplement adds prediction-backed success/failure cases for each of the six datasets and separates them from source-publication images in Fig. 1. Those images illustrate resource appearance, not model evidence. Full S pose/input records are absent from the upstream snapshot, so a verified, clip-matched pose–input–output panel cannot yet be reconstructed. We do not substitute an unrelated frame or invented feature description. The release records the missing assets needed to complete that reviewer request.

4.5. Release and Limitations

The independent project repository provides six-dataset prediction logs, scoring/plotting code, the actual runner and SFT configuration, a table-to-file map, checksums, and source-access/license notes. It regenerates the shared LLM point metrics and figures from committed outputs. Source poses, complete train/validation/test input layers, adapters, and training logs are not included; full training/inference reproduction requires these assets from their authorized source. Dataset licenses are not replaced by a blanket repository license. Original RGB recordings are not redistributed, and GMFCS public access is to de-identified trajectories rather than clinical video.

Small test populations, performed-gait provenance, source simulation uncertainty, unequal class support, and unresolved training-artifact availability limit interpretation. The subject-disjoint S contract addresses the *design* of the old GMFCS overlap concern, but its full split inputs must still be recovered for independent verification; no newly executed leakage-free rerun is claimed. These boundaries are disclosed with the reproducible result checks instead of treating missing evidence as a successful experiment.

5. Conclusion

We introduced GaitEval, a six-dataset benchmark for grounding dataset-specified gait-label predictions in pose-derived motion evidence. Controlled inputs and same-feature linear references distinguish limited signal from failed evidence use. The five-family and same-model results retain the accepted finding of limited zero-shot agreement in many tested settings, with dataset–representation exceptions under the shared six-dataset protocol. Task-specific adaptation recovers selected in-distribution mappings; the retained ATAXIA/PATHOGAIT adaptation results do not establish clinical or population-level superiority. Reliable transfer is not observed in the tested bidirectional ATAXIA–PATHOGAIT probe. Overall, GaitEval shows that LLM grounding in pose-derived gait evidence is heterogeneous and representation-dependent: adaptation can recover dataset-specific mappings in distribution, but reliability beyond the tested settings remains unresolved.

References

[1] Jacquelin Perry. *Gait Analysis: Normal and Pathological Function*. SLACK Incorporated, 1992. 1, 2

[2] Richard Baker. Gait analysis methods in rehabilitation. *Journal of NeuroEngineering and Rehabilitation*, 3:4, 2006. 2

[3] Alberto Muro-de-la Herran, Begonya Garcia-Zapirain, and Amaia Mendez-Zorrilla. Gait analysis methods: An overview of wearable and non-wearable systems, highlighting clinical applications. *Sensors*, 14(2):3362–3394, 2014. 1

[4] Zhe Cao, Tomas Simon, Shih-En Wei, and Yaser Sheikh. Realtime multi-person 2D pose estimation using part affinity fields. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7291–7299, 2017. 1, 3

[5] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.

[6] Valentin Bazarevsky, Ivan Grishchenko, Karthik Raveendran, Tyler Zhu, Fan Zhang, and Matthias Grundmann. BlazePose: On-device real-time body pose tracking. *arXiv preprint arXiv:2006.10204*, 2020. 1

[7] Jan Stenum, Kendra M. Cherry-Allen, Connor O. Pyles, Rachel D. Reetzke, Michael F. Vignos, and Ryan T. Roemmich. Applications of pose estimation in human health and performance across the lifespan. *Sensors*, 21(21):7315, 2021. 1, 2

[8] Jan Stenum, Melody M. Hsu, Alexander Y. Pantelyat, and Ryan T. Roemmich. Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change. *PLOS Digital Health*, 3(3):e0000467, 2024. 1, 2

[9] David A. Winter. *Biomechanics and Motor Control of Human Movement*. John Wiley & Sons, 4th edition, 2009. 1, 3

[10] Michael H. Schwartz and Adam Rozumalski. The gait deviation index: A new comprehensive index of gait pathology. *Gait & Posture*, 28(3):351–357, 2008. 1, 3

[11] Hisham Khalil, Ahmed Mohamed Saad Emam Saad, and Uswah Khairuddin. Diagnosis of cerebellar ataxia based on gait analysis using human pose estimation: A deep learning approach. In *2022 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES)*, pages 201–206, 2022. 1, 2

[12] Daniel Deng, Jill L. Ostrem, Vy Nguyen, Daniel D. Cummins, Julia Sun, Anupam Pathak, Simon Little, and Reza Abbasi-Asl. Interpretable video-based tracking and quantification of parkinsonism clinical motor states. *npj Parkinson's Disease*, 10:122, 2024. 1, 3

[13] Łukasz Kidziński, Bryan Yang, Jennifer L. Hicks, Apoorva Rajagopal, Scott L. Delp, and Michael H. Schwartz. Deep neural networks enable quantitative movement analysis using single-camera videos. *Nature Communications*, 11:4054, 2020. 1, 2

[14] Mandy Lu, Kathleen L. Poston, Adolf Pfefferbaum, Edith V. Sullivan, Li Fei-Fei, Kilian M. Pohl, Juan Carlos Niebles, and Ehsan Adeli. Vision-based estimation of MDS-UPDRS gait scores for assessing parkinson's disease motor severity. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 637–647, 2020. 1

[15] Kooksung Jun, Yongwoo Lee, Sanghyub Lee, Deok-Won Lee, and Mun Sang Kim. Pathological gait classification using Kinect v2 and gated recurrent neural networks. *IEEE Access*, 8:139881–139891, 2020. 1, 2, 3

[16] Juri Opitz and Sebastian Burst. Macro F1 and Macro F1. *arXiv preprint arXiv:1911.03347*, 2019. 1, 5

[17] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009. 1, 5

[18] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1877–1901, 2020. 1, 4

[19] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. 1, 4

[20] Sheldon R. Simon. Quantification of human motion: Gait analysis—benefits and limitations to its application to clinical problems. *Journal of Biomechanics*, 37(12):1869–1880, 2004. 2

[21] Alexander Mathis, Pranav Mamidanna, Kevin M. Cury, Taiga Abe, Venkatesh N. Murthy, Mackenzie Weygandt Mathis, and Matthias Bethge. DeepLabCut: Markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience*, 21(9):1281–1289, 2018. 2

[22] Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10):e1011462, 2023. 2

[23] Tanmay T. Verlekar, Henri De Vroey, Kurt Claeys, Hans Hallez, Luís D. Soares, and Paulo L. Correia. Estimation and validation of temporal gait features using a markerless 2D video system. *Computer Methods and Programs in Biomedicine*, 175:45–51, 2019. 2

[24] Masataka Yamamoto, Koji Shimatani, Masaki Hasegawa, Yuichi Kurita, Yuto Ishige, and Hiroshi Takemura. Accuracy of temporo-spatial and lower limb joint kinematics parameters using OpenPose for various gait patterns with orthosis. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 29:2666–2675, 2021.

[25] Andrea Sabo, Carolina Gorodetsky, Alfonso Fasano, Andrea Iaboni, and Babak Taati. Concurrent validity of Zeno instrumented walkway and video-based gait features in adults with parkinson's disease. *IEEE Journal of Translational Engineering in Health and Medicine*, 10:2100511, 2022. 2

[26] Sijie Yan, Yuanjun Xiong, and Dahua Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI)*, pages 7444–7452, 2018. 2

[27] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. Two-stream adaptive graph convolutional networks for skeleton-based action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12026–12035, 2019.

[28] Ziyu Liu, Hongwen Zhang, Zhenghao Chen, Zhiyong Wang, and Wanli Ouyang. Disentangling and unifying graph convolutions for skeleton-based action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 143–152, 2020. 2

[29] Elham Dolatabadi, Babak Taati, and Alex Mihailidis. An automated classification of pathological gait using unobtrusive sensing technology. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 25(12):2336–2346, 2017. 2

[30] Tanmay T. Verlekar, Luís D. Soares, and Paulo L. Correia. Automatic classification of gait impairments using a markerless 2D video-based system. *Sensors*, 18(9):2743, 2018.

[31] Javier Ortells, María Trinidad Herrero-Ezquerro, and Ramón A. Mollineda. Vision-based gait impairment analysis for aided diagnosis. *Medical & Biological Engineering & Computing*, 56(9):1553–1564, 2018. 2

[32] Jung Hwan Shin, Ri Yu, Jed Noel Ong, Chan Young Lee, Seung Ho Jeon, Hwanpil Park, Han Joon Kim, Jehee Lee, and Beomseok Jeon. Quantitative gait analysis using a pose-estimation algorithm with a single 2D-Video of Parkinson's Disease patients. *Journal of Parkinson's Disease*, 11(3):1271–1283, 2021. 2

[33] Andrea Sabo, Sina Mehdizadeh, Kimberley-Dale Ng, Andrea Iaboni, and Babak Taati. Assessment of Parkinsonian Gait in older adults with dementia via human pose tracking in video data. *Journal of NeuroEngineering and Rehabilitation*, 17:97, 2020.

[34] Qingyi Zeng, Peipei Liu, Ningbo Yu, Jialing Wu, Weiguang Huo, and Jianda Han. Video-based quantification of gait impairments in Parkinson's Disease using skeleton–silhouette fusion convolution network. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:2912–2922, 2023. 2

[35] Evelina Pantzar-Castilla, Andrea Cereatti, Giulio Figari, Nicolò Valeri, Gabriele Paolini, Ugo Della Croce, Anders Magnuson, and Jacques Riad. Knee joint sagittal plane movement in cerebral palsy: A comparative study of 2-dimensional markerless video and 3-dimensional gait analysis. *Acta Orthopaedica*, 89(6):656–661, 2018. 2

[36] Luca Lonini, Yaejin Moon, Kyle Embry, R. James Cotton, Kelly McKenzie, Sophia Jenz, and Arun Jayaraman. Video-based pose estimation for gait analysis in stroke survivors during clinical assessments: A proof-of-concept study. *Digital Biomarkers*, 6(1):9–18, 2022. 2

[37] Wasifur Rahman, Masum Hasan, Md Saiful Islam, Titilayo Olubajo, Jeet Thaker, Abdelrahman R. Abdelkader, Phillip Yang, Henry Paulson, Gulin Oz, Alexandra Durr, Thomas Klockgether, Tetsuo Ashizawa, READISCA Investigators, and Ehsan Hoque. Auto-Gait: Automatic ataxia risk assessment with computer vision from gait task videos. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(1):26:1–26:19, 2023. 2, 3

[38] Katsuki Eguchi, Hiroaki Yaguchi, Hisashi Uwatoko, Yuki Iida, Shinsuke Hamada, Sanae Honma, Asako Takei, Fumio Moriwaka, and Ichiro Yabe. Feasibility of differentiating gait in Parkinson's Disease and spinocerebellar degeneration using a pose estimation algorithm in two-dimensional video. *Journal of the Neurological Sciences*, 464:123158, 2024. 3

[39] Rahm Ranjan, David Ahmedt-Aristizabal, Mohammad Ali Armin, and Juno Kim. Computer vision for clinical gait analysis: A gait abnormality video dataset. *arXiv preprint arXiv:2407.04190*, 2024. 3

[40] Tanmay T. Verlekar, Paulo L. Correia, and Luís D. Soares. Using transfer learning for classification of gait pathologies. In *2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 2376–2381, 2018. 3

[41] Pedro Albuquerque, Tanmay Tulsidas Verlekar, Paulo Lobato Correia, and Luís Ducla Soares. A spatiotemporal deep learning approach for automatic pathological gait classification. *Sensors*, 21(18):6202, 2021. 3

[42] Vida Adeli, Ivan Klabučar, Javad Rajabi, Benjamin Filtjens, Soroush Mehraban, Diwei Wang, Hyewon Seo, Trung-Hieu Hoang, Minh N. Do, Candice Muller, Claudia Neves de Oliveira, Daniel Boari Coelho, Pieter Ginis, Moran Gilat, Alice Nieuwboer, et al. CARE-PD: A multi-site anonymized clinical dataset for Parkinson's Disease gait assessment. In *Advances in Neural Information Processing Systems*, 2025. Datasets and Benchmarks Track. 3

[43] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. MotionGPT: Human motion as a foreign language. In *Advances in Neural Information Processing Systems*, pages 20067–20079, 2023. 3

[44] Ling-Hao Chen, Shunlin Lu, Ailing Zeng, Hao Zhang, Benyou Wang, Ruimao Zhang, and Lei Zhang. MotionLLM: Understanding human behaviors from human motions and videos. *arXiv preprint arXiv:2405.20340*, 2024.

[45] Jiawei Mo, Yixuan Chen, Rifen Lin, Yongkang Ni, Min Zeng, Xiping Hu, and Min Li. MoChat: Joints-grouped spatio-temporal grounding LLM for multi-turn motion comprehension and description. *arXiv preprint arXiv:2410.11404*, 2024. 3

[46] Shaopeng Yang, Jilong Wang, Saihui Hou, Xu Liu, Chunshui Cao, Liang Wang, and Yongzhen Huang. Bridging gait recognition and large language models sequence modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3460–3469, 2025. 3

[47] Diwei Wang, Cédric Bobenrieth, and Hyewon Seo. AGIR: Assessing 3D gait impairment with reasoning based on LLMs. *arXiv preprint arXiv:2503.18141*, 2025. 3

[48] Ruize Yang, Ann Kennedy, and R. James Cotton. BiomechGPT: Towards a biomechanically fluent multimodal foundation model for clinically relevant motion tasks. *arXiv preprint arXiv:2505.18465v1*, 2025. 3

[49] Erdong Chen, Yuyang Ji, Jacob K. Greenberg, Benjamin Steel, Faraz Arkam, Abigail Lewis, Pranay Singh, and Feng Liu. BioGait-VLM: A tri-modal vision-language-biomechanics framework for interpretable clinical gait assessment. *arXiv preprint arXiv:2603.08564*, 2026. 3

[50] Carlo Dindorf, Jonas Dully, Rebecca Keilhauer, Michael Lorenz, and Michael Fröhlich. Evaluating large language models for gait classification using text-encoded kinematic waveforms. *arXiv preprint arXiv:2603.13317*, 2026. 3

[51] Xi Fang, Weijie Xu, Fiona Anting Tan, Jiani Zhang, Ziqing Hu, Yanjun Qi, Scott Nickleach, Diego Socolinsky, Srinivasan Sengamedu, and Christos Faloutsos. Large language models (LLMs) on tabular data: Prediction, generation, and understanding—a survey. *Transactions on Machine Learning Research*, 2024. 3

[52] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems*, pages 34892–34916, 2023. 3

[53] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. A survey on multimodal large language models. *National Science Review*, 11(12):nwae403, 2024. 3

[54] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu. Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(7):1325–1339, 2014. 3

[55] Robert Palisano, Peter Rosenbaum, Stephen Walter, Dianne Russell, Ellen Wood, and Barbara Galuppi. Development and reliability of a system to classify gross motor function in children with cerebral palsy. *Developmental Medicine & Child Neurology*, 39(4):214–223, 1997. 3

[56] Hisham Khalil, Ahmed Saad, and Uswah Khairuddin. Dataset for gait analysis of cerebellar ataxic patients and healthy adults using MediaPipe pose. Mendeley Data, 2023. Version 1. 3

[57] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. 4

[58] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[59] Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025. 4

[60] Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. ChatGLM: A family of large language models from GLM-130B to GLM-4 all tools. *arXiv preprint arXiv:2406.12793*, 2024. 4

[61] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645:633–638, 2025. 4

[62] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 24824–24837, 2022. 4

[63] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, 1995. 4

[64] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. 4

[65] Jacob Cohen. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220, 1968. 5

[66] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993. 5

[67] Ahmed A. Al-Jubouri, Israa Hadi Ali, and Yasen Rajihy. Gait and full body movement dataset of autistic children classified by rough set classifier. *Journal of Physics: Conference Series*, 1818(1):012201, 2021. 3